ICT2113 · Data Structures and Algorithms

Revision Guide and Practice

Original synthetic reading for demonstration and independent study. It is not an official syllabus, lecture note, or assessment specification.

Learning objectives

After this reading, a learner should be able to explain the central idea, apply it to a small example, and identify at least one assumption or limitation.

1. Main idea

Review invariants, operation costs, and preconditions before choosing an algorithm. Big-O describes growth as input size increases; it does not by itself measure implementation constants or memory locality.

2. Example and reasoning

Practice: compare array and linked-list insertion; prove binary search maintains a valid interval; trace BFS on a small graph; count comparisons in insertion sort; identify the invariant that makes a binary search tree searchable.

3. A method for practice

1. State the problem and define the terms that affect the answer.
2. Write down the assumptions and information available.
3. Work through a small example and make each step visible.
4. Check the result against a second case or independent explanation.
5. Summarize what the result supports and what it does not establish.

4. Questions for study

- Explain “revision guide and practice” in your own words and give a new example.
- Which assumption matters most in the example, and how would the answer change if it were false?
- Describe one common mistake and a practical way to detect it.
- Write a short checklist another learner could use to review this work.

Reflection

Separate evidence from assumptions. If a conclusion depends on information not provided here, identify what would need to be verified.